

ROZHEN OBSERVATORY, Bulgaria

WMO: 157260

Lat: 41.6950N

Lon: 24.7390E

Elev: 5741

StdP: 11.89

Time Zone: 2.00 (EUE)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.3	7.4	-5.7	5.0	20.4	-1.7	6.2	18.4	18.9	19.1	17.6	22.1	8.6	320	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	74.7	56.6	71.9	56.2	69.3	55.4	59.9	68.8	58.3	67.3	57.0	66.1	6.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.7	85.1	63.0	55.2	80.3	61.1	53.7	75.9	59.9	29.2	68.9	27.9	67.2	27.0	66.0	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	14.5	12.2	DB	-0.3	80.5	5.4	3.9	-4.1	83.3	-7.3	85.6	-10.3	87.8	-14.2	90.6	(4)
(5)			WB	-0.8	63.6	5.3	1.6	-4.6	64.8	-7.7	65.7	-10.7	66.7	-14.6	67.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	25.7	25.8	31.7	38.8	48.4	54.8	58.9	59.7	51.6	44.7	37.4	28.7	(6)
(7)		DBStd	14.51	9.57	9.84	8.69	7.26	6.98	6.66	5.82	5.64	6.96	8.52	9.77	9.38	(7)
(8)		HDD50	3813	753	679	569	344	116	27	5	3	62	203	391	661	(8)
(9)		HDD65	8330	1218	1099	1034	786	515	309	203	180	402	629	829	1126	(9)
(10)		CDD50	994	0	0	1	8	66	172	282	304	110	39	12	0	(10)
(11)		CDD65	38	0	0	0	0	0	5	15	17	1	0	0	0	(11)
(12)		CDH74	136	0	0	0	0	0	16	65	54	1	0	0	0	(12)
(13)		CDH80	13	0	0	0	0	0	2	9	2	0	0	0	0	(13)
(14)	Wind	WSAvg	6.0	6.3	6.3	6.7	6.1	5.9	5.9	5.8	5.6	5.4	5.5	6.3	6.0	(14)
(15)	Precipitation	PrecAvg	28.1	2.7	2.4	2.5	2.2	2.7	2.4	1.9	1.4	2.0	2.2	2.4	3.3	(15)
(16)		PrecMax	42.0	5.9	5.9	5.2	5.1	5.7	5.4	5.1	3.8	5.6	5.6	5.0	6.7	(16)
(17)		PrecMin	17.0	0.4	0.6	0.7	0.7	1.5	0.9	0.2	0.3	0.4	0.3	0.3	0.3	(17)
(18)		PrecStd	5.1	1.3	1.4	1.2	0.9	0.9	1.0	1.1	0.9	1.2	1.3	1.1	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	50.5	56.0	63.7	68.7	75.5	80.5	79.6	72.7	67.8	63.2	51.9	(19)
(20)			MCWB	36.0	37.3	41.5	49.5	52.0	58.7	59.3	57.6	53.3	50.8	46.2	41.0	(20)
(21)		2%	DB	44.3	44.2	51.0	56.6	64.9	71.2	75.0	74.6	69.0	62.5	57.8	46.8	(21)
(22)			MCWB	33.4	34.1	39.1	43.5	51.0	56.6	56.4	56.9	53.1	49.2	44.2	37.7	(22)
(23)		5%	DB	40.2	41.0	46.6	52.4	61.9	68.1	71.7	71.9	65.0	58.8	52.8	43.2	(23)
(24)			MCWB	32.5	33.1	37.2	42.3	49.9	55.3	56.3	56.7	51.7	47.5	40.8	37.3	(24)
(25)		10%	DB	37.0	37.7	43.0	49.0	58.6	65.1	68.5	68.9	61.4	55.3	49.1	40.5	(25)
(26)			MCWB	32.3	32.4	35.6	40.3	48.4	54.3	55.7	55.4	50.7	46.0	39.5	34.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.7	43.1	50.3	54.9	61.2	62.6	61.2	56.4	53.7	49.9	43.7	(27)
(28)			MCD B	42.8	45.2	53.1	63.1	63.6	69.9	74.6	70.0	66.4	64.8	59.2	46.8	(28)
(29)		2%	WB	36.4	36.2	40.3	45.6	52.5	58.7	60.0	59.5	54.5	51.0	46.4	41.1	(29)
(30)			MCD B	40.3	40.6	48.2	54.3	61.0	67.4	69.2	68.6	64.1	58.5	52.4	44.3	(30)
(31)		5%	WB	34.3	34.7	38.3	43.2	50.9	57.0	58.2	58.0	53.0	49.5	44.0	38.5	(31)
(32)			MCD B	38.1	39.0	44.9	50.6	59.0	65.5	66.7	67.8	61.6	56.0	49.2	41.7	(32)
(33)		10%	WB	32.6	33.1	36.3	41.3	49.4	55.3	56.8	56.8	51.8	47.9	41.9	36.2	(33)
(34)			MCD B	35.8	36.9	41.9	47.2	56.8	62.4	65.3	66.7	59.0	53.3	46.4	39.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	11.0	12.2	12.0	12.3	12.5	13.6	13.6	12.6	11.8	11.4	9.7	(35)
(36)			MCDBR	14.2	15.0	15.6	16.3	15.9	15.5	17.0	16.3	16.6	15.9	15.0	12.7	(36)
(37)			MCWBR	8.8	9.4	8.8	9.2	8.9	7.7	7.7	7.2	8.4	9.3	9.5	8.2	(37)
(38)		5% WB	MCD BR	11.4	12.1	14.0	14.7	14.4	14.2	14.6	15.0	14.6	13.5	12.2	11.4	(38)
(39)			MCWBR	8.6	8.3	8.5	8.9	8.9	8.4	8.3	8.0	8.0	8.6	8.8	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.304	0.345	0.393	0.384	0.385	0.382	0.388	0.368	0.331	0.302	0.282	(40)	
(41)		taud	2.458	2.392	2.303	2.160	2.251	2.272	2.281	2.260	2.302	2.411	2.468	2.458	(41)	
(42)		Ebn at Noon	281	287	285	278	283	282	281	276	274	273	267	268	(42)	
(43)		Edh at Noon	24	30	37	46	43	42	41	41	36	29	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	552	811	1139	1459	1780	1945	2061	1886	1404	964	639	488	(44)	
(45)		RadStd	74	99	104	161	127	169	150	125	106	100	60	72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.81	N/A	N/A	N/A	+0.94	+0.90	-148	-265	+133	N/A

Nomenclature: See separate page